

Formula-first co-design of a 2-DOF PIDF for a wide-range non-ideal Buck converter: LQI weights as the search space, an exact structural realization, and switched-model validation

Anonymous author(s)

Affiliation withheld for review

Abstract

Placeholder.

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1. Introduction

Digitally controlled DC–DC converters are routinely tuned by metaheuristics: particle swarms, genetic algorithms, bee colonies and grey-wolf variants search the gains of a PI, PID or filtered PID (PIDF) against a tracking index [18, 26, 28]. Two weaknesses recur in that literature, and both are visible to an industrial reader. First, the searched object is a vector of gains with no structural meaning, so a good gain vector carries no stability or robustness certificate and the search space is sized by guesswork. Second, the evidence is usually produced on the same averaged model that drives the search, so switching ripple, duty saturation and discontinuous conduction, which decide whether a tuning is admissible on the real power stage, never enter the comparison [23, 15].

This paper addresses both points on one concrete, audited plant: a non-ideal asynchronous Buck converter supplied from a fixed 48 V source and commanded over the whole attainable range 1 V to 46 V. The design is *formula first*. The searched object is not a gain vector but the set of Bryson excursions that define a linear–quadratic–integral (LQI) cost; every candidate

is solved from the algebraic Riccati equation and then realized *exactly* as a two-degree-of-freedom PIDF (2-DOF PIDF) that reads the output voltage only. Every evaluated point is therefore a genuine LQI law, with the classical LQR return-difference guarantees, and at the same time a controller that a practitioner can implement in a standard PID block with setpoint weights.

The central theoretical observation is stronger than a transfer-function identity. We show that the derivative filter of the realizing PIDF, when its pole is placed at the equivalent-series-resistance (ESR) zero $N_C = 1/(r_C C)$ and its state is initialized consistently, reconstructs the capacitor voltage *exactly*, whatever happens in the inductor branch. Consequently the output-feedback PIDF and the full-state LQI generate identical trajectories under duty saturation, anti-windup activity, discontinuous conduction and any mismatch of L , r_L , R_{on} , r_d or V_f ; only an unmeasured load current, a mismatch of the output network (R, r_C, C) , or sampling break the equivalence. This turns a nominal, small-signal equivalence into a structural statement that can be checked on a simulator to machine precision, and it tells the practitioner exactly which sensors the PIDF saves and at what price.

The optimization layer compares six equal-budget global operators—PSO, GA, ABC and three project variants pAEABC, pIGWO and pIGWO-DLH—each followed by a common Gaussian-process refinement stage (refineBO), over 30 matched seeds. The seven algorithms are not the contribution by themselves; the question they answer is whether the *LQI-weight parameterization* helps them. We therefore run the identical protocol on five other controller families that share the evaluator, the budget and the specifications: the full-state twin of the design (same weights, measured i_L), the same 2-DOF PIDF searched directly in gain space, a PIDF with free filter pole and setpoint weights, a PID and a PI. Six fixed-rule comparators—PI, PID, PIDF (Type-III rule), LQR, LQG and LQI—complete the picture, and every frozen controller is replayed on a switched replica of the reference Simulink/Simscape model on the complete 24→1→46 V mission, with a closed energy balance.

A previous version of this protocol accepted candidates on an averaged model guarded by an analytical switching-ripple bound, and four of six branch winners failed on the switched model. We reproduce that failure mode, show that it is caused by coexisting switching limit cycles whose selection depends on the preceding transient, and replace the analytical bound by three *measured* inequalities obtained from a 34 ms switched transition sequence inside each evaluation, a multi-fidelity constraint that roughly dou-

bles the cost of a candidate. Feasibility is also stated in absolute engineering units (overshoot, load deviation, device current, steady error) instead of ratios to a reference controller, so that every family is judged against the same specification.

The contributions are:

1. a formula-first chain from the non-ideal converter equations to the exact output-feedback realization of any LQI law as a 2-DOF PIDF, with all analytical quantities (equilibria, transfer functions, Riccati gain, LQR margins, a common- P D-region certificate over the operating polytope) instantiated and independently re-computed;
2. a structural equivalence theorem (Proposition 2) showing that the PIDF filter state equals the capacitor voltage along every trajectory of the plant, with a proof, its exact domain of validity and a machine-precision numerical confirmation;
3. a constrained co-design problem in the space of LQI weights with 33 explicit inequalities in engineering units, three of which are measured on the switched model during the search;
4. a matched, equal-budget comparison of seven algorithms across six controller families, which isolates the value of the LQI parameterization from the value of the optimizer;
5. a switched-model validation on the reference mission against PI, PID, PIDF, LQR, LQG and LQI, reported with the specifications each controller meets or violates rather than with a single aggregate score.

Section 3 derives the model and describes the switched replica. Section 4 states the LQI synthesis and the equivalence results. Section 5 defines the co-design problem and the algorithms. Section 6 reports the results and Section 7 concludes. All code, including a MATLAB S-function that runs the same control layer in the reference Simulink model, is released with the paper.

2. Related work and positioning

Model-based converter control. State-space averaging supplies the standard low-cost converter model [23, 15]; non-ideal averaged plants with inductor and capacitor ESR, switch resistance and diode drop have been used to tune implementable low-order controllers with hardware validation [31], and robust nonlinear or observer-based laws extend disturbance rejection [1, 32, 7].

91 Convex methods—MPC solved as a quadratic program [22], LMI regional
 92 pole placement [9]—give certificates relative to their model class but do not
 93 make fixed-structure PIDF tuning convex once saturation, discontinuous con-
 94 duction and event-wise requirements are included.

95 *LQ design and PID structure.* The LQR return-difference inequality guaran-
 96 tees an infinite upward gain margin, a gain reduction margin of one half and
 97 60° of phase margin for the full-state loop [19, 29, 3]; the guarantee is not
 98 inherited by observer-based implementations [13], and loop-transfer recovery
 99 trades it against observer bandwidth [14]. Bryson’s rule converts admissible
 100 excursions into diagonal weights [8]. Setpoint weighting and back-calculation
 101 anti-windup are the standard means of shaping the command response of a
 102 PID and of handling actuator saturation [5, 4, 6]. The connection exploited
 103 here—that the full-state LQI law of a second-order plant with an output-
 104 network zero is a filtered PID of the measured output—is known in nominal
 105 transfer-function form; what is new is its trajectory-level validity and the use
 106 of the Riccati weights, not the gains, as the search space.

107 *Metaheuristic tuning.* PSO [21], genetic algorithms, ABC [20] and GWO
 108 [24] with improved variants [16, 25, 2] are widely used to tune converter con-
 109 trollers [18, 26, 35]; surrogate-based Bayesian optimization is attractive when
 110 evaluations are expensive [33, 10]. Constraint handling by Deb’s feasibility
 111 rules avoids penalty weights [11], and matched-seed non-parametric statistics
 112 are the accepted way to compare stochastic solvers [12]. A weighted scalar
 113 index does not produce a Pareto front [28]; we therefore report a constrained
 114 scalar problem and do not claim a multi-objective characterization.

115 *Switching effects in digital control.* In digitally controlled PWM converters,
 116 the sampled switching ripple and the duty resolution can sustain limit cycles
 117 that an averaged model does not predict [27]. This is the mechanism behind
 118 the rejection of averaged-feasible candidates reported in Section 6.2, and the
 119 reason why two inequalities of the present problem are measured on the
 120 switched model.

121 Table 1 positions the study. The gap is not a new metaheuristic but the
 122 conjunction of (i) a structurally meaningful search space whose every point
 123 carries an LQ certificate and an exact output-feedback realization, (ii) a fea-
 124 sibility set stated in engineering units and partly measured on the switched
 125 plant, (iii) matched equal-budget comparisons that separate the parameteri-

Table 1: Positioning with respect to representative Buck control and tuning studies. “Not reported” refers to the stated scope of each work only.

Study	Plant evidence	Controller / search	Relative to the present question
Siddhartha et al. [31]	non-ideal Buck, dSPACE	IMC-PID, direct	no repeated solver comparison, no switched energy audit
Izci et al. [18]	averaged Buck	PID, AEO + Nelder-Mead	gain-space search, no structural certificate, no switched admissibility
Nanyan et al. [26]	averaged Buck	PID, improved SCA	gain-space search, single fidelity
Siano and Citro [30]	Buck prototype	fuzzy, MO-PSO	no equal-budget solver comparison
Abdelmalek et al. [1]	Buck, experimental	observer-based nonlinear, PSO	tracking/robustness focus, not a matched co-design
Coutinho et al. [10]	multiloop process	PID, Bayesian optimization	not a power converter
Present work	audited non-ideal Buck, 1–46 V, switched replica with energy balance	LQI weights $\rightarrow$ exact 2-DOF PIDF; 6 global operators + refineBO; 6 families	structural equivalence theorem; 33 inequalities (2 measured on the switched model); switched validation vs. PI, PID, PIDF, LQR, LQG, LQI

zation from the optimizer, and (iv) a closed-energy-balance switched validation on the full commanded range.

3. Non-ideal Buck converter: from the circuit to the switched replica

3.1. Circuit and conventions

Fig. 1 shows the asynchronous converter. The state is $x = [i_L \ v_C]^\top$; i_L flows from the switching node to the output, i_C flows downward in the capacitor branch and the load disturbance i_p is positive when it draws additional current. The source is fixed, $V_{in} = 48$ V. The high-side switch is an IRF540N represented by its static on-resistance R_{on} ; the freewheel diode follows the affine conduction law of the STPST10H100SF data sheet, $v_D = V_f + r_d i_D$ [17, 34]. Table 2 lists the values, all decoded from the workspace of the reference Simulink model.

3.2. Averaged model and loss-aware equilibrium

The output node and the ESR give

$$i_L = i_C + \frac{v_o}{R} + i_p, \quad v_o = v_C + r_C i_C, \quad (1)$$

hence

$$i_C = \frac{R(i_L - i_p) - v_C}{R + r_C}, \quad v_o = \frac{R}{R + r_C} v_C + \frac{R r_C}{R + r_C} (i_L - i_p). \quad (2)$$

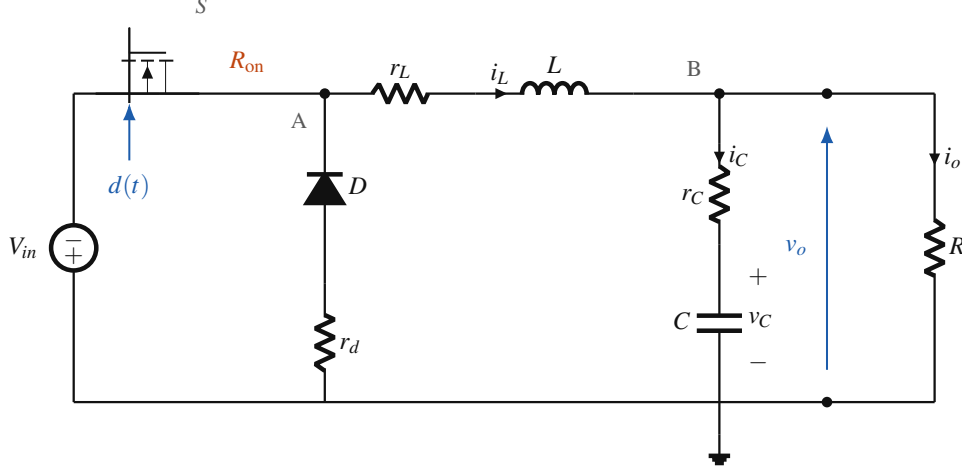


Figure 1: Non-ideal asynchronous Buck converter and adopted conventions.

142 With the switch on, $L\dot{i}_L = V_{in} - (r_L + R_{on})i_L - v_o$; with the diode conducting,
 143 $L\dot{i}_L = -V_f - (r_L + r_d)i_L - v_o$; in both cases $C\dot{v}_C = i_C$. Duty averaging yields

$$L\dot{i}_L = dV_{in} - [r_L + dR_{on} + (1-d)r_d]i_L - (1-d)V_f - v_o. \quad (3)$$

144 At equilibrium $I_L = V_o/R + I_p$ and the loss-aware duty is

$$d_{ff} = \frac{V_o + (r_L + r_d)I_L + V_f}{V_{in} - (R_{on} - r_d)I_L + V_f}. \quad (4)$$

145 Every controller of this paper uses (4) with nominal parameters and $I_p = 0$
 146 as feedforward; the feedforward never sees the simulated plant.

147 3.3. Small-signal model and transfer functions

148 Let $\gamma = R/(R + r_C)$, $\rho_0 = r_L + r_d + d_0(R_{on} - r_d)$ and $E_d = V_{in} +$
 149 $V_f - (R_{on} - r_d)I_{L,0}$. Linearizing (2)–(3) gives $\delta\dot{x} = A\delta x + B_d\delta d + B_p\delta i_p$,
 150 $\delta v_o = C_y\delta x + D_p\delta i_p$ with

$$A = \begin{bmatrix} -(\rho_0 + \gamma r_C)/L & -\gamma/L \\ \gamma/C & -\gamma/(RC) \end{bmatrix}, \quad B_d = \begin{bmatrix} E_d/L \\ 0 \end{bmatrix}, \quad C_y = [\gamma r_C \quad \gamma], \quad (5)$$

Table 2: Electrical and numerical parameters (reference model workspace).

Quantity	Value	Role
V_{in}	48 V, constant	source
L, r_L	100 μ H, 0.10 Ω	inductor branch
C, r_C	100 μ F, 0.05 Ω	capacitor branch
R	10 Ω	nominal load
R_{on}	0.05 Ω	IRF540N static surrogate
V_f, r_d	0.42 V, 0.020 Ω	STPST10H100SF affine law
R_s ($C_s = \infty$)	100 k Ω	resistive snubber across the switch
$R_{d,body}$	0.01 Ω	MOSFET body diode
f_s	50 kHz	PWM carrier
T_s	10 μ s	controller sampling ($= 1/(2f_s)$)
$[d_{min}, d_{max}]$	[0.02, 0.98]	total duty saturation

Table 3: Equilibria, monic $\Delta(s)$, G_{vd} numerator and stationary conduction screen at the three commanded levels (nominal load, $i_p = 0$). $\omega_z = 1/(r_C C) = 2 \times 10^5$ rad/s at every level.

V_o (V)	$I_{L,o}$ (A)	d_0	ρ_0 (Ω)	E_d (V)	$\Delta(s)/(LC(R+r_C))$	G_{vd} numerator	$\Delta I_{L,pp}$ (A)	I_{crit} (A)	$\bar{I}_L - I_{crit}$ (A)	R_{crit}/R
1	0.1	0.029576	0.120887	48.417	$s^2 + 2701.41s + 1.00705 \times 10^8$	$2.40881 \times 10^4 s + 4.81761 \times 10^9$	0.1941	0.0970	0.0030	1.030
24	2.4	0.511045	0.135331	48.348	$s^2 + 2845.85s + 1.00849 \times 10^8$	$2.40537 \times 10^4 s + 4.81075 \times 10^9$	2.3470	1.1735	1.2265	2.045
46	4.6	0.972868	0.149186	48.282	$s^2 + 2984.40s + 1.00987 \times 10^8$	$2.40209 \times 10^4 s + 4.80418 \times 10^9$	0.2496	0.1248	4.4752	36.856

151 $B_p = [\gamma r_C/L \quad -\gamma/C]^\top$ and $D_p = -\gamma r_C$. The control-to-output transfer
152 function is

$$G_{vd}(s) = \frac{E_d R(1 + r_C C s)}{\Delta(s)}, \quad (6)$$

$$\Delta(s) = LC(R + r_C)s^2 + [L + C((R + r_C)\rho_0 + R r_C)]s + R + \rho_0,$$

153 with the ESR zero $\omega_z = N_C = 1/(r_C C) = 2 \times 10^5$ rad/s. Table 3 instantiates
154 (4)–(6) at the three commanded levels; the closed form and a state-space
155 conversion agree to machine precision. At the 24 V synthesis point, (A, B_d)
156 and (A, C_y) have rank two and the integral-augmented pair has rank three.

157 3.4. Conduction mode across the range

158 For a duty d_{ff} the inductor ripple, the boundary current and the bound-
159 ary load are

$$\Delta I_{L,pp} = \frac{V_o(1 - d_{ff})}{L f_s}, \quad I_{crit} = \frac{\Delta I_{L,pp}}{2}, \quad R_{crit} = \frac{2L f_s}{1 - d_{ff}}. \quad (7)$$

160 The stationary screen (Table 3) is positive at every level but not uniform:
 161 at 1 V the current margin is 2.96 mA (3.05 % of I_{crit}) and R_{crit} is only 1.030
 162 times the nominal load. The switched model is more pessimistic than this
 163 screen: over the steady 1 V window it spends 15.00 % of the time in discon-
 164 tinuous conduction at 60.57 % efficiency, because the diode drop is 42 % of
 165 the commanded output. Discontinuous conduction is thus a nominal regime
 166 of the mission, and every controller is validated in it.

167 3.5. Switched replica of the reference model

168 The acceptance model (Fig. 2) is the Simulink/Simscape Electrical model
 169 `buck_fo_lqi_fixed_reference_v6_1_46V_R2022a.slx`. For the campaign we
 170 use a replica written from its block parameters (`cmo/sim.py`): a resistive
 171 switch with snubber R_s and a body diode $R_{d,body}$; the affine freewheel diode,
 172 which blocks reverse current so that discontinuous conduction arises by it-
 173 self (the 1 ns snubber-leakage branch is solved quasi-statically); a continuous
 174 sawtooth carrier whose edges are located exactly inside the integration step;
 175 a controller sampled at T_s with a one-sample output delay, exactly as the
 176 S-function of the reference model. The step is 0.2 μ s (100 steps per switching
 177 period) with fourth-order Runge–Kutta integration.

178 The energy boundary is the one declared in the model workspace:

$$\begin{aligned} P_{in} &= V_{in}i_{in}, & P_u &= \frac{v_o^2}{R} + v_o i_p, \\ P_{loss} &= r_L i_L^2 + r_C i_C^2 + v_{SW} i_{SW} + v_D i_D, \end{aligned} \quad (8)$$

179 the snubber dissipation being contained in $v_{SW} i_{SW}$. The three energies are
 180 integrated with the same Runge–Kutta stages as the states, so the residual
 181 $E_{in} - E_u - E_{loss} - \Delta E_{stored}$ measures integration consistency only. It stays
 182 below 4.7×10^{-6} % of the input energy for every replay reported here.

183 The replica reproduces the controller-independent figures of the reference
 184 model: 15.00 % of discontinuous conduction and 60.57 % efficiency in the 1 V
 185 window (15.27 % and 60.51 % in the Simulink audit of the previous version
 186 of this study). Its discrete LQI comparator gain, synthesized by the rule
 187 of Section 5.5, equals the gain stored in the model workspace to all printed
 188 digits, and the fixed PI and LQI comparators reproduce the Simulink over-
 189 shoot and peak current to within 2 % (Section 6.2). A MATLAB S-function
 190 (`matlab/cmo_sfun.m`) implements the same control layer line by line, and
 191 `matlab/run_cmo_validation.m` replays every frozen controller on the refer-
 192 ence `.slx` without modifying it.

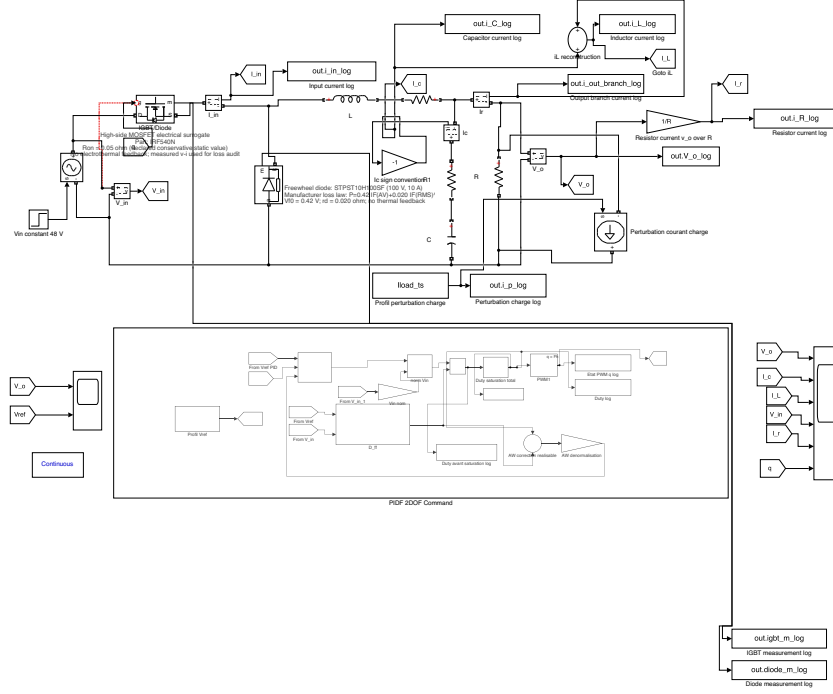


Figure 2: Reference Simulink/Simscape Electrical switching model (fixed 48 V source, IRF540N surrogate, STPST10H100SF diode, resistive snubber, continuous PWM at 50 kHz, sampled controller subsystem and energy loggers). Only the controller S-function is replaced when a candidate is replaced.

193 4. LQI synthesis in weight space and its exact 2-DOF PIDF real- 194 ization

195 4.1. Integral augmentation and Bryson weights

196 Let $z(t) = \int_0^t (r - v_o) d\tau$. Around the synthesis equilibrium ($V_o = 24$ V)
197 the augmented plant is

$$\dot{x}_a = A_a x_a + B_a \delta d, \quad A_a = \begin{bmatrix} A & 0 \\ -C_y & 0 \end{bmatrix}, \quad B_a = \begin{bmatrix} B_d \\ 0 \end{bmatrix}, \quad x_a = \begin{bmatrix} \delta x \\ z \end{bmatrix}, \quad (9)$$

198 and the LQI law $\delta d = -K x_a$, $K = [K_{ix} \ K_{vx} \ K_z]$, minimizes $\int_0^\infty (x_a^\top Q x_a +$
199 $R_d \delta d^2) dt$ with

$$A_a^\top P + P A_a - P B_a R_d^{-1} B_a^\top P + Q = 0, \quad K = R_d^{-1} B_a^\top P. \quad (10)$$

200 Bryson's rule assigns $Q = \text{diag}(\Delta_i^{-2}, \Delta_v^{-2}, (\Delta_v/\omega_B)^{-2})$ and $R_d = \Delta_d^{-2}$: Δ_i
 201 and Δ_v are admissible current and voltage excursions and the integral state,
 202 in Vs, is scaled by a target bandwidth ω_B . The duty excursion is fixed at
 203 its physical value $\Delta_d = (d_{\max} - d_{\min})/2 = 0.48$, which removes the scale
 204 redundancy $(Q, R_d) \rightarrow (\alpha Q, \alpha R_d)$ of (10). The decision vector of the co-
 205 design is

$$\xi = [\log_{10} \Delta_i, \log_{10} \Delta_v, \log_{10} \omega_B, \log_{10} K_b]^\top, \quad (11)$$

206 where K_b is the back-calculation rate of the anti-windup, which lies outside
 207 the linear theory and is searched alongside the weights. The normalization
 208 record ξ_0 uses $\Delta_i = 15$ A, $\Delta_v = 46$ V (the largest commanded output) and
 209 $\omega_B = K_b = 2\pi \cdot 1500$ rad/s.

210 4.2. Frequency-domain realization

211 For $i_p = 0$ the output network gives, in the Laplace domain, $v_C = v_o/(1 +$
 212 $r_C Cs)$ and $i_L = v_o/R + C s v_o/(1 + r_C Cs)$. Substituting into the state part of
 213 the LQI law,

$$\frac{K_{ix} i_L + K_{vx} v_C}{v_o} = \frac{K_{ix}}{R} + \frac{K_{ix} Cs + K_{vx}}{1 + r_C Cs} = K_p + K_d \frac{N_C s}{s + N_C}, \quad (12)$$

214 which exposes one constant and one filtered derivative term.

215 **Proposition 1** (Nominal realization). *The output-error compensator gener-*
 216 *ated by the full-state LQI law is the 2-DOF PIDF*

$$d = d_{ff}(r) + K_p(br - v_o) + \frac{K_i}{s}(r - v_o) + K_d \frac{N_C s}{s + N_C}(cr - v_o), \quad (13)$$

217 *with*

$$\begin{aligned} K_p &= K_{vx} + \frac{K_{ix}}{R}, & K_i &= -K_z, & K_d &= C(K_{ix} - r_C K_{vx}), \\ N &= N_C = \frac{1}{r_C C}, & b &= 1, & c &= 0. \end{aligned} \quad (14)$$

218 *Proof.* Split the right-hand side of (12) into its zero-frequency value $K_{vx} +$
 219 K_{ix}/R and the remainder $C(K_{ix} - r_C K_{vx})N_C s/(s + N_C)$. The equilibrium
 220 map $x_{eq}(r) = [r/R \ r]^\top$ contributes $+(K_{ix}/R + K_{vx})r = K_p r$, hence $b = 1$;
 221 the state-feedback law has no derivative of the reference, hence $c = 0$; the
 222 integral state contributes $-K_z z$, hence $K_i = -K_z$. $\square$

Because $N_C = 2 \times 10^5 \text{ rad/s} < \pi/T_s = 3.14 \times 10^5 \text{ rad/s}$, the exact filter pole is implementable at the adopted sampling period, and no projected surrogate is needed: every candidate of the search is realized with its own exact pole.

4.3. Structural (trajectory-level) equivalence

Proposition 1 is a statement about transfer functions, i.e. about the linear nominal plant with consistent initialization. The next result shows that the realization is in fact exact along every trajectory of the *nonlinear* converter, because it only uses the output network, never the inductor branch. Write the PIDF of (13) in state form,

$$\begin{aligned} \tilde{d} &= d_{ff}(r) + K_p(br - v_o) + x_I + K_d N(cr - v_o - x_F), & d &= \text{sat}(\tilde{d}), \\ \dot{x}_I &= K_i(r - v_o) + K_b(d - \tilde{d}), & \dot{x}_F &= N(cr - v_o - x_F), \end{aligned} \quad (15)$$

and the LQI with the same feedforward and anti-windup as

$$\begin{aligned} \tilde{d} &= d_{ff}(r) - K_{ix}(i_L - \frac{r}{R}) - K_{vx}(v_C - r) - K_z z, \\ \dot{z} &= (r - v_o) + \frac{K_b}{|K_z|}(d - \tilde{d}). \end{aligned} \quad (16)$$

Proposition 2 (Structural equivalence). *Let r be piecewise constant, let the PIDF use (14) with $c = 0$, $N = N_C$ computed from the true (r_C, C) , and let $x_F(0) = -v_C(0)$ and $x_I(0) = -K_z z(0)$. Then, along every trajectory of the converter—averaged or switched, in continuous or discontinuous conduction, with or without active saturation and anti-windup, and for any values of L , r_L , R_{on} , r_d , V_f and V_{in} :*

- (i) $x_F(t) = -v_C(t)$ for all $t \geq 0$;
- (ii) the two unsaturated commands satisfy

$$\tilde{d}_{LQI}(t) - \tilde{d}_{PIDF}(t) = -K_{ix}\left(i_o(t) - \frac{v_o(t)}{R}\right), \quad (17)$$

where $i_o = i_L - i_C$ is the actual output-branch current and R the nominal load;

- (iii) if $i_o \equiv v_o/R$ (no load disturbance, nominal load), the two closed loops generate identical trajectories.

246 *Proof.* (i) The capacitor branch gives $v_o = v_C + r_C C \dot{v}_C$ exactly, whatever
 247 flows into the output node, so $\dot{v}_C = N_C(v_o - v_C)$. Let $\varepsilon = x_F + v_C$. Between
 248 reference steps, with $c = 0$, $\dot{\varepsilon} = N_C(-v_o - x_F) + N_C(v_o - v_C) = -N_C\varepsilon$, and
 249 ε is continuous across steps because $c = 0$. Hence $\varepsilon(0) = 0$ implies $\varepsilon \equiv 0$.
 250 (ii) With (i), the derivative term of (15) is $K_d N_C(v_C - v_o) = -K_d N_C r_C \dot{v}_C =$
 251 $-(K_{ix} - r_C K_{vx})i_C$. In (16) substitute $i_L = i_o + i_C$ and $v_C = v_o - r_C i_C$:
 252 $-K_{ix}(i_L - r/R) - K_{vx}(v_C - r) = K_p(r - v_o) - (K_{ix} - r_C K_{vx})i_C - K_{ix}(i_o - v_o/R)$.
 253 With $b = 1$ and $x_I = -K_z z$ the difference of the two commands is (17). (iii) If
 254 the right-hand side of (17) vanishes, both laws issue the same $\tilde{d}$ and d at every
 255 instant. Then $\frac{d}{dt}(x_I + K_z z) = K_i(r - v_o) + K_b(d - \tilde{d}) - |K_z|(r - v_o) - K_b(d - \tilde{d}) =$
 256 0 because $K_z < 0$, so $x_I = -K_z z$ is preserved, and the two closed-loop
 257 systems coincide; uniqueness of solutions gives identical trajectories. $\square$

258 Three practical consequences follow. First, the PIDF does not approxi-
 259 mate the LQI on the nonlinear converter; it *is* the LQI whenever the load
 260 current is the nominal one, and the saturation, anti-windup and conduction-
 261 mode effects that invalidate small-signal arguments do not break it. Second,
 262 the price of removing the current sensor is stated exactly by (17): the full-
 263 state law contains an instantaneous proportional feedback $-K_{ix}(i_o - v_o/R)$
 264 on the unmodelled output current, which the PIDF must recover through its
 265 integral action. Third, the equivalence is lost by exactly three mechanisms:
 266 a load disturbance or load mismatch (through (17)), a mismatch of (r_C, C)
 267 (through $N \neq N_C$ in (i)), and sampling, because the discrete filter recon-
 268 structs v_C only between samples while the sampled LQI reads i_L with its
 269 switching ripple. Section 6.1 verifies each case.

270 4.4. Guarantees and their scope

271 *LQR margins.* For the full-state augmented loop $L_K(s) = K(sI - A_a)^{-1}B_a$
 272 the return-difference identity gives $|1 + L_K(j\omega)| \geq 1$, hence the gain margin
 273 $(1/2, \infty)$ and a phase margin of at least 60° [29, 3]. By Proposition 2(iii)
 274 these margins are those of the PIDF loop at the input of the nominal linear
 275 plant; they do not survive sampling or an observer [13].

276 *Operating-polytope certificate.* The augmented closed-loop matrix is affine
 277 in $(\rho_0(d_0), E_d(I_{L,0}))$. Over $v_o \in [1, 46]$ V and $i_p \in [0, 0.5]$ A the rectangle
 278 $d_0 \in [0.0295764, 0.9744132]$, $I_{L,0} \in [0.1, 5.1]$ A has four vertices $A_{cl,k}$. A

279 common $P \succ 0$ with

$$A_{cl,k}P + PA_{cl,k}^\top + 2\alpha P \prec 0, \quad \begin{bmatrix} \sin \theta(M_k + M_k^\top) & \cos \theta(M_k - M_k^\top) \\ \cos \theta(M_k^\top - M_k) & \sin \theta(M_k + M_k^\top) \end{bmatrix} \prec 0, \quad (18)$$

280 $M_k = A_{cl,k}P$, $\zeta = \cos \theta$, places every closed-loop matrix of the convex hull
 281 in $\{\Re s < -\alpha\} \cap \{\zeta(s) > \zeta\}$ [9]. Bisection gives the boundary values; a 1 %
 282 safety factor is applied and the recovered P is re-checked for strictness.

283 *Discrete implementation.* All controllers run at T_s with a one-sample com-
 284 putational delay, total duty saturation and forward-Euler integral and back-
 285 calculation updates. Proposition 2(i) says that the derivative-filter state is an
 286 observer of the capacitor voltage, $x_F = cr - v_C$, and it should be discretized as
 287 one. The usual zero-order-hold discretization, $x_F[k+1] = \phi x_F[k] + (1-\phi)w[k]$
 288 with $w = cr - v_o$ and $\phi = e^{-NT_s}$, uses at sample k an estimate built from
 289 $v_o[k-1]$ only; during a 45 V slew v_o moves by about 2.7 V per sample and
 290 $K_d N \approx 0.7$ turns this lag into a duty error of order one. We therefore use
 291 the triangle-hold (first-order hold) update, exact when v_o is piecewise linear
 292 between samples,

$$x_F[k] = \phi x_F[k-1] + (1-\phi)w[k-1] + \gamma(w[k] - w[k-1]), \quad \gamma = 1 - \frac{1-\phi}{NT_s}, \quad (19)$$

293 evaluated with the current sample before the command $\tilde{d}[k] = d_{ff} + K_p(br[k] -$
 294 $v_o[k]) + x_I[k] + K_d N(w[k] - x_F[k])$ is formed, and initialized with $x_F[0] =$
 295 $cr[0] - v_C(0)$. Section 6.1 shows that this single choice decides whether the
 296 sampled PIDF remains equivalent to the sampled LQI.

297 4.5. Numerical instantiation

298 At the 24 V point, (5) evaluates to

$$A = \begin{bmatrix} -1850.8259 & -9950.2488 \\ 9950.2488 & -995.0249 \end{bmatrix} \text{s}^{-1}, \quad (20)$$

$$B_d = \begin{bmatrix} 483480 \\ 0 \end{bmatrix}, \quad C_y = [0.0497512 \quad 0.9950249].$$

299 For ξ_0 , $Q = \text{diag}(0.00444444, 0.00047259, 41978.4686)$, $R_d = 4.34027778$
 300 and (10) gives $K = [0.03564341 \quad 0.01212576 \quad -98.34550916]$ with a rel-
 301 ative Riccati residual of 2.18×10^{-14} and closed-loop poles -4243.01 and

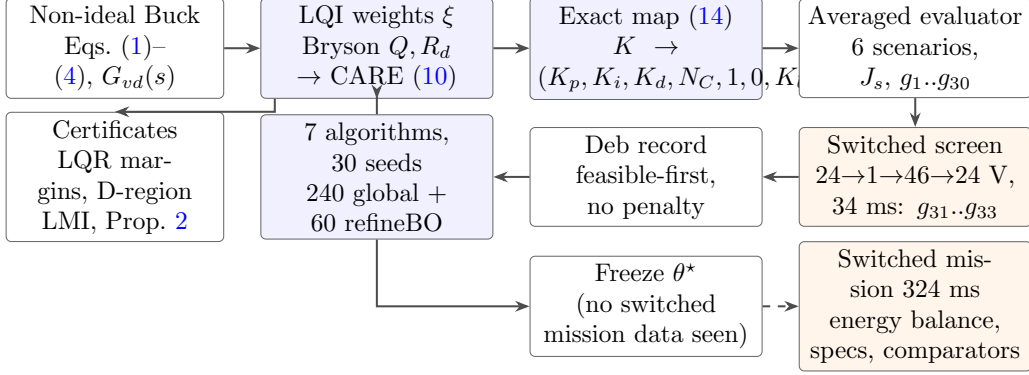


Figure 3: Formula-first CMO–LQI–PIDF workflow. Solid arrows: evaluation inside the search; dashed arrow: acceptance replay after the freeze. The switched screen is the only switched information used during the search.

302 $-7917.86 \pm j6986.57$ rad/s. The realization (14) gives

$$\begin{aligned}
 \theta_0 &= [K_p, K_i, K_d, N, b, c, K_b] \\
 &= [0.0156901028, 98.3455092, 3.5037119 \times 10^{-6}, 2 \times 10^5, 1, 0, 9424.778],
 \end{aligned}
 \tag{21}$$

303 with a zero coefficient residual and a maximum relative frequency-domain
 304 error of 8.67×10^{-16} over $[10^{-1}, 10^8]$ rad/s. The minimum return difference is
 305 1.000000 and the phase margin of the full-state loop is 87.04° . The operating-
 306 polytope certificate at ξ_0 is $\alpha_{cert} = 4157.9286$ rad/s and $\zeta_{cert} = 0.7330911$
 307 ($\lambda_{\min}(P) > 0$ and strictly negative LMI eigenvalues were re-verified on the
 308 recovered matrices).

309 5. Constrained co-design problem and algorithms

310 5.1. Workflow

311 Fig. 3 summarizes the protocol. Offline, a candidate ξ is mapped by (10)
 312 and (14) to a 2-DOF PIDF; the averaged evaluator runs six scenarios and
 313 the switched screen runs three short steady windows; Deb ordering ranks the
 314 record. After 240 global and 60 local calls the best record of each run is
 315 frozen, and only then replayed on the full switched mission, which is never
 316 visible to the search.

Table 4: Averaged scenarios (the source is 48 V in all of them).

ID	T (ms)	reference	load pulse (+0.5 A)
S0	90	24→1→46 V at 30, 60 ms	15–25 ms
S1a	40	24→1 V at 10 ms	–
S1b	40	1→46 V at 10 ms	–
S1c	40	46→24 V at 10 ms	–
S2a	40	24 V	10–20 ms
S2b	40	46 V	10–20 ms

5.2. Scenarios and objective

The averaged evaluator integrates (3) with the hybrid guard $i_L \geq 0$ (RK4, 2 μ s) under the sampled control layer of Section 4. Six scenarios are declared (Table 4); each starts from the analytical equilibrium preceding its first event. For scenario s the seven criteria are the loss intensity E_{loss}/E_u , IAE, ISE, global ITAE, event-reset ITAE, RMSE and MAE, each divided by its value for the normalization record ξ_0 . The scalar objective is the mean of the 6×7 ratios,

$$J_s(\xi) = \frac{1}{42} \sum_{s=1}^6 \mathbf{1}^\top r_s(\xi), \quad J_s(\xi_0) = 1. \quad (22)$$

MAE is IAE divided by the horizon and RMSE is the square root of ISE divided by it, so two of the seven entries are redundant; the induced weights (2/7 on the IAE family, 2/7 on the ISE family, 1/7 on each other criterion) are declared rather than left implicit. The objective ranks feasible candidates only.

330 *5.3. Thirty-three inequalities in engineering units*

331 For each scenario five inequalities $g_{s,j} \leq 0$ are evaluated separately:

$$\begin{aligned}
g_{s,1} &= \frac{\max_t \sigma_k (v_o - r_k)}{1.5 \text{ V}} - 1 && \text{overshoot beyond each new level,} \\
g_{s,2} &= \frac{\max_{t \in \mathcal{T}_p} |r - v_o|}{0.5 \text{ V}} - 1 && \text{load-pulse deviation,} \\
g_{s,3} &= \frac{\max_t i_L + \Delta I_{L,pp}^{\max}/2}{23 \text{ A}} - 1 && \text{device current,} \\
g_{s,4} &= \frac{\max_k |\bar{e}_k|}{25 \text{ mV}} - 1 && \text{steady error,} \\
g_{s,5} &= \frac{\rho_{\ell,s} - 1}{0.005} - 1, \quad \rho_{\ell,s} = \frac{E_{\text{loss}}/E_u}{(E_{\text{loss}}/E_u)_{\xi_0}} && \text{loss intensity,}
\end{aligned} \tag{23}$$

332 where σ_k is the sign of the k -th reference step, $\mathcal{T}_p$ runs from the onset of
333 the load pulse to the next event, $\bar{e}_k$ is the mean error over the last 2 ms of
334 each constant segment and 23 A is the IFR540N continuous drain rating at
335 100 °C. The averaged peak current is augmented by the worst half ripple
336 $\Delta I_{L,pp}^{\max}/2 = 1.17 \text{ A}$ (Table 3). A family that does not apply to a scenario (no
337 step, no load pulse) is set to -1 . The loss-intensity gate keeps the design
338 loss-aware: no scenario may convert energy less efficiently, per useful joule,
339 than the Bryson record ξ_0 by more than 0.5 %. These 30 inequalities are
340 completed by three measured ones,

$$g_{31} = \frac{\max_k \text{pp}(d)}{0.20} - 1, \quad g_{32} = \frac{\max_k |\bar{e}_{sw,k}|}{25 \text{ mV}} - 1, \quad g_{33} = \frac{\max_t i_{L,sw}}{23 \text{ A}} - 1, \tag{24}$$

341 obtained from one run of the switched replica over the transition sequence
342 $24 \rightarrow 1 \rightarrow 46 \rightarrow 24 \text{ V}$ (steps at 2, 14 and 24 ms, 34 ms in total, 1 μs RK4 with
343 exact PWM edges). The duty peak-to-peak and the mean error are measured
344 over the last 4 ms of each level. A candidate is feasible when $\sum_j \max(g_j, 0) \leq$
345 10^{-4} ; the screen costs about 15 ms per candidate against 14 ms for the six
346 averaged scenarios.

347 *Why the screen contains large transitions.* The protocol was frozen after two
348 pilot campaigns that are reported, not hidden. Pilot 1 used the analytical rip-
349 ple bound of the earlier version of this study, $\rho_{rip} = \max_V |C_{fb}(z=-1)| \Delta V_{o,pp}(V)/2 \leq$
350 0.20, where C_{fb} is the discrete feedback path evaluated at the controller
351 Nyquist frequency, which coincides with the carrier frequency because $T_s =$

352 $1/(2f_s)$. Every branch winner it produced satisfied the bound ($\rho_{rip} \approx 0.07$ –
 353 0.19) and yet, on the switched model, several sustained a limit cycle with
 354 duty swings of 0.7 – 0.9 at 24 V and offsets of 0.3 – 1.8 V. Pilot 2 replaced the
 355 bound by steady 10 ms switched windows started *from equilibrium* at 1 , 24
 356 and 46 V. Its best design (178 of 180 runs feasible) passed that screen, but on
 357 the reference mission it entered, after the large $24 \rightarrow 1$ V transition, a different
 358 limit cycle in which one sample in four clips at $d_{\min}$, leaving a -42 mV offset
 359 at 1 V. The loop thus holds *coexisting* switching limit cycles, and the one
 360 reached depends on the transient that precedes it; this is why the final screen
 361 is a transition sequence. ρ_{rip} is kept as a reported diagnostic only. Both pi-
 362 lots used the zero-order-hold filter; the triangle-hold update (19), adopted
 363 after the equivalence audit of Section 6.1, removes most of the limit-cycling
 364 designs by itself.

365 *Useful-energy band.* The earlier protocol also required $|E_u/E_{u,\xi_0} - 1| \leq 0.5\%$
 366 in every scenario. We report this quantity but no longer constrain it. An-
 367 chored to ξ_0 , it penalizes the removal of the 9.8 V undershoot of ξ_0 on the
 368 $46 \rightarrow 24$ V step, because a controller that holds 24 V delivers more energy than
 369 one that undershoots; anchored to the ideally regulated task, it penalizes
 370 every controller by 1 – 5% on step-down scenarios, where the asynchronous
 371 converter can only discharge through its load. With absolute overshoot, load-
 372 deviation and steady-error gates the delivered task is already pinned, and the
 373 loss intensity is normalized per useful joule, so under-delivery cannot improve
 374 it.

375 Feasibility is stated in absolute units, not as non-inferiority against ξ_0 .
 376 With ratio gates, 24 of 36 constraints are active at ξ_0 and a gentler normaliza-
 377 tion record leaves no candidate that dominates it; with absolute gates every
 378 controller family of Section 5.5 is judged against the same specification, and
 379 ξ_0 itself is infeasible because it overshoots by 6.2 V on the $1 \rightarrow 46$ V step.

380 5.4. Algorithms and budgets

381 Constraint handling uses Deb’s rules [11]: feasible beats infeasible, fea-
 382 sible records are ranked by J_s , infeasible ones by aggregate violation; no
 383 penalty weight is tuned. All methods work in the normalized box $u \in [0, 1]^n$,
 384 receive the matched seed $2026091200 + k$, $k = 1, \dots, 30$, and the same initial
 385 population of 12: ξ_0 first, six further individuals within a normalized radius
 386 0.05 of it, five uniform. One counted call is one complete evaluation (six

Table 5: Algorithms 1–7. Population 12, 240 global calls, then 60 refineBO calls initialized from the same run’s history (300 calls per run).

Alg.	Exploration / update	Exploitation / survival	Fixed parameters
1 PSO [21]	velocity with independent cognitive and social draws	personal and global Deb incumbents	$w = 0.9 - 0.5\tau$, $c_1 = c_2 = 1.7$, $ v \leq 0.2$
2 GA	binary Deb tournaments, SBX crossover	polynomial mutation, elitist ($\mu + \lambda$)	$\eta_c = 15$, $\eta_m = 20$, $p_m = 1/n$
3 ABC [20]	one-coordinate differential move (employed, onlooker)	greedy Deb replacement, scout	limit $\max(5, \text{round}(0.5 n_p n))$
4 pAEABC	ABC move scaled by diversity $s_k = \text{clip}(0.5 + 0.15/D_k, 0.5, 2)$, extra coordinate kick w.p. $0.25(1 - \tau)$	as ABC	inspired by Alsamia et al. [2]
5 pIGWO	three-leader GWO with $a = 2(1 - \tau^2)$, Gaussian kick w.p. $0.15(1 - \tau) + 0.02$	elitist pooling	inspired by Hou et al. [16]
6 pIGWO–DLH	linear $a = 2 - 2k/K$ GWO trial and an independently evaluated radial local-hunt trial	better trial replaces the wolf (non-elitist)	$R_i = \ x_i - x_\alpha\ /2$; inspired by Nadimi-Shahraki et al. [25]
7 refineBO	shared-kernel Matérn-5/2 GPs on J_s and $g_1, \dots, g_{33}$; pool of 1000 around the 10 Deb-best records, radius $0.04 \rightarrow 0.003$	$\alpha = P_f(\text{EI} + 0.02\sigma_J) + 10^{-4}P_f$, $P_f = \prod_j \Phi(-\mu_j/\sigma_j)$; restoration mode if no feasible incumbent	60 calls

averaged scenarios, the switched screen, 33 inequalities). Table 5 lists the operators.

The prefix “p” marks project variants that are inspired by, but not claimed to be operator-identical with, the cited algorithms. τ is the fraction of the budget consumed. The product P_f of marginal feasibility probabilities is an independence approximation used for ordering only; every accepted point is checked by the exact evaluator. Wall time is logged but never interpreted as an energy or performance quantity.

5.5. Controller families and fixed-rule comparators

To separate the value of the parameterization from that of the optimizer, the identical protocol (evaluator, 33 inequalities, 7 algorithms, 30 seeds, 300 calls) is run on six families:

- **CMO–LQI–PIDF**: ξ of (11) $\rightarrow$ (10) $\rightarrow$ (14); 4 variables, reads v_o only;
- **CMO–LQI–SF**: the same ξ and the same continuous gain, applied as full-state LQI with measured i_L and i_C (the twin of Proposition 2);
- **PIDF-direct**: the same 2-DOF structure ($b = 1$, $c = 0$, $N = N_C$) with $(\log K_p, \log K_i, \log K_d, \log K_b)$ searched directly over the gain ranges spanned by the LQI box;

- 406 • **PIDF7-direct**: $K_p, K_i, K_d, N, b, c, K_b$ all free (7 variables);
- 407 • **PID-direct**: backward-difference derivative on $-v_o$, $b = 1$, (K_p, K_i, K_d, K_b)
- 408 free;
- 409 • **PI-direct**: (K_p, K_i, K_b) free.

410 *Fixed-rule comparators.* Six classical designs share the control layer and are
 411 synthesized at 24 V, $T_s = 10 \mu\text{s}$, with the delay $1.5T_s$ (computation plus
 412 ZOH) in every frequency-domain rule. *PI*: crossover $0.03f_s$, phase margin
 413 60° , $K_b = K_i/K_p$. *PID*: zeros cancelling the *LC* pair of $\Delta(s)$, crossover set for
 414 60° of phase margin including the ESR zero, backward-difference derivative
 415 on $-v_o$, $b = 1$, $c = 0$, $K_b = 1/\sqrt{T_i T_d}$. *PIDF*: the Type-III rule, i.e. the
 416 PID zeros plus a filter pole on the ESR zero $N = N_C$, 60° of phase margin.
 417 *LQR*: discrete Riccati solution on the ZOH model with the Bryson limits of
 418 ξ_0 , measured i_L and $v_C = v_o - r_C i_C$, no integral action. *LQG*: the LQR gain
 419 with a steady-state Kalman predictor on v_o only (process noise 0.01 in duty,
 420 measurement noise 0.05 V). *LQI*: discrete Riccati solution of the integral-
 421 augmented ZOH model with the same weights, measured state, $K_b = \omega_B$.
 422 None of them is tuned by the proposed search; they represent what the
 423 standard rules deliver on this plant.

424 5.6. Statistics and validation

425 Methods are compared as matched blocks over the 30 seeds with the
 426 Friedman test and Holm-adjusted paired Wilcoxon signed-rank tests [12];
 427 a run's score is J_s if feasible and 10^3 plus its violation otherwise, which is
 428 consistent with Deb ordering. Families are compared pairwise on the same
 429 (method, seed) cells. The retained controller of each family is the best feasible
 430 record over its 180 runs; it is frozen and replayed on the 324 ms reference
 431 mission of the Simulink model (24→1→46 V at 108 and 216 ms, +0.5 A load
 432 pulse at 54–90 ms, initial state at the 24 V equilibrium), with the energy
 433 balance of (8).

434 6. Results

435 6.1. E_q

436 6.2. Sw

437 7. Conclusions

438 Appendix A. Reproducibility record

439 *Reference model.* buck_fo_lqi_fixed_reference_v6_1.46V_R2022a.slx, SHA-
440 256 948fbae2679b57ed726a92eeebcae398aaf0ca6b9b3a74216759ca80b390c7fe.

441 All plant, mission and logging parameters of Table 2 were decoded from its
442 model workspace; the snubber is resistive ($R_s = 100\text{ k}\Omega$, $C_s = \infty$) on the
443 MOSFET only, the diode snubber is disabled, and the MOSFET body diode
444 has $R_d = 0.01\text{ }\Omega$, $V_{fd} = 0$. The saved model initializes L at 1.2 A and C at
445 12 V; all replays of this paper start at the 24 V equilibrium (2.4 A, 24 V).

446 *Commands.* From the repository root:

```
447 python scripts/cmo_formula_first.py    # Sections 3-4, Prop. 2 tests
448 python scripts/cmo_campaign.py         # 6 families x 6 methods x 30 seeds
449 python scripts/cmo_analyze.py          # statistics, freeze, switched replay
450 python scripts/cmo_figures.py; python scripts/cmo_numbers.py
```

451 In MATLAB (R2022a or later, Simscape Electrical, Control System Tool-
452 box):

```
453 cmo_formula_first
454 T = run_cmo_validation(slx, 'results/cmo/controllers.csv');
```

455 The second call replaces only the controller S-function (`cmo_sfun`, a line-
456 by-line port of `cmo/ctrl.py`) and never saves the model.

457 *Seeds and budgets.* Seeds 2026091201, ..., 2026091230; population 12; 240
458 global and 60 refineBO calls per run; 30 runs per method and family. Each
459 checkpoint stores the method, seed, best decision vector, objective, the 33
460 inequalities, the global-stage record and the best-so-far curve.

461 Appendix B. Protocol changes with respect to the earlier version

462 Data availability

463 All code (Python replica, evaluator, algorithms, analysis, figures), the
464 MATLAB S-function and replay script for the reference Simulink model,
465 the per-run checkpoints and the generated tables are archived with the
466 manuscript repository; a public DOI will be assigned at acceptance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table B.6: Declared changes and their reasons.

Change	Reason
Feasibility in engineering units instead of ratios to ξ_0	with ratio gates 24 of 36 constraints are active at ξ_0 , and a gentle record leaves no dominating candidate; absolute gates make all families comparable
Analytical ripple bound $\rightarrow$ switched transition screen (g_{31} – g_{33})	pilots 1 and 2: the bound and a steady screen both admitted limit-cycling designs (Section 5)
Device current limit 23 A (IRF540N at 100 °C)	the earlier retained design used 90 % of the 25 °C rating
refineBO budget stated once: 60 calls	the earlier text gave both 20 and 60
Structural equivalence (Prop. 2)	replaces the claim that saturation, DCM and PWM break the realization
Triangle-hold (FOH) discretization of the derivative filter, Eq. (19)	the ZOH filter makes the sampled PIDF deviate by up to 14.54 V from the sampled LQI on the 1 $\rightarrow$ 46 V step; FOH reduces it to 0.35 V (identified by the equivalence audit, before the final campaign)
Useful-energy band reported, not constrained	ill-posed on step-down transients of an asynchronous converter (Section 5); identified before the final campaign
Six added families and six fixed-rule comparators on one specification	separates parameterization from optimizer